National University of Computer & Emerging Sciences, Karachi

Fall-2025 School of Computing (BSCS, BSSE, BSCY, BSAI)

Assignment # 2

Subject: Database Systems -CS2005	**Post Date: 8/10/2025**
Total Marks: 50	**Due Date:** 24/10/2025
Course Instructors: Dr. Zulfiqar, Dr. Farrukh Salim, Basit Jasani, Ali Naseer, Atiya Jokhio, Javeria Farooq, Hajira Ahmed, Huzaifa, Umair	

Instructions to be strictly followed.

- For all questions involving SQL Queries:
    - ***Submit the SQL Scripts in a .txt file.***
- It should be obvious that submitting your work after the due date will result in zero points being awarded.
- Plagiarism (copying/cheating) and late submissions result in a zero mark.

Question #1: A film studio needs a database to track its **Films**, the **Actors** they hire, and the **Production Companies** they work with.

Business Rules:

1. **Films**: Every **Film** has a unique FilmID, a Title, a Budget, and a ReleaseDate.
2. **Actors**: Every **Actor** has a unique ActorID, a Name, and a Phone number.
3. **Casting**: A **Film** casts many **Actors**, and an **Actor** can be cast in many **Films**. The role an actor plays (CharacterName) must be recorded when they are cast in a film.
4. **Production**: Each **Film** is made by exactly one **Director** (who is not tracked separately as an entity). The director's Name is stored directly in the FILM record.
5. **Funding**: A **Film** is funded by one or more **Production Companies** (CompanyID, Name). A company can fund many films.

A. Identify all primary Entities and their corresponding Attributes from the scenario. Underline the Primary Key (PK) for each entity.

B. Identify the three main Relationships (e.g., A relates to B). State the Cardinality (1:1, 1:M, M:N) for each relationship.

C. Rule 3 describes a Many-to-Many (M:N) relationship. Name the Associative Entity (Linking Table) needed to resolve it and list all the attributes it will contain (including the PK).

D. Explain why DirectorName (Rule 4) should NOT be made into a separate entity in this simplified scenario.

Question #2: Design an ER Schema for keeping track of information about an Airline Reservation System, given the description below:

There are 6 different airlines in 6 different countries: Canada – AirCan, USA - USAir, UK - BritAir, France - AirFrance, Germany - LuftAir, Italy - ItalAir. Their flights involve the following 12 cities: Toronto and Montreal in Canada, New York and Chicago in US, London and Edinburgh in UK, Paris and Nice in France, Bonn and Berlin in Germany, Rome and Naples in Italy.

In each of the 12 cities, there is a (single) booking office. You are going to design a central air-reservation database to be used by all booking offices.

The flight has a unique flight number, airline code, business class indicator, smoking allowed indicator. Flight availability has flight number, date + time of departure, number of total seats available in business class, number of booked seats in business class, number of total seats available in economy class, and number of booked seats in economy class.

The customers may come from any country, not just the 6 above, and from any province/state, and from any city. Customer has first & last name, mailing address, zero or more phone numbers, zero or more fax numbers, and zero or more email addresses. Mailing address has street, city, province or state, postal code and country. Phone/fax number has country code, area code and local number. Email address has only one string, and no structure is assumed. A customer can book one or more flights. Two or more customers may have same mailing address and/or same phone number(s) and/or same fax number(s). But the email address is unique for each customer. First and last names do not have to be unique.

Booking has a unique booking number, booking city, booking date, flight number, date + time of departure (in local time, and time is always in hours and minutes), date + time of arrival (in local time), class indicator, total price (airport tax in origin + airport tax in destination + flight price – in local currency. The flightprice for business class is 1.5 times of the listed flight price), status indicator (three types: booked, Canceled – the customer canceled the booking, scratched – the customer had not paid in full 30 days prior to the departure), customer who is responsible for payment, amount-paid-so far (in local currency), outstanding balance (in local currency), the first & last names to be printed on the ticket. The airport taxes must be stored in local currencies (i.e. Canadian dollars, US dollars, British Pounds, French francs, German marks, and Italian Liras). Since the exchange rates change daily, they also must be stored for calculations of all prices involved. Though France, Germany, and Italy have had a common currency for a while, we used the names of their original currencies to involve in this exercise currency exchange rates and their changes.

Question #3: A company called **EduTemps** provides temporary teaching staff to schools throughout England. The following unnormalized table lists information about the temporary teachers, the hours they worked, and the schools they worked at.

NIN	contractNo	hours	tName	schoolID	schoolName	schoolCity
2103	T9001	10	Brown A	S301	Park School	Manchester
2198	T9001	18	Khan R	S301	Park School	Manchester
2205	T9001	20	Lewis H	S114	Green School	Birmingham
2103	T9001	12	Brown A	S114	Green School	Birmingham

a. **The table shown above is susceptible to update anomalies. Provide examples of insertion, deletion, and update anomalies.**
b. **Describe and illustrate the process of normalizing the table to 3NF, by identifying the functional dependencies represented by the attributes. State any assumptions you make.**

Question#4: The casting office created the following un-normalized table, FILM_CASTING, to track casting details.

FILM_CASTING

FilmID, ActorID, FilmTitle, DirectorName, ActorName, AgentID, AgentPhone

Functional Dependencies (FDs) are given as:

1. FilmID→FilmTitle, DirectorName
2. ActorID→ActorName, AgentID
3. AgentID→AgentPhone
4. FilmID, ActorID→All other attributes (No other single attribute determines the whole record).

A. Identify the Primary Key (PK) for the FILM_CASTING table based on the given Functional Dependencies.
B. The table is currently in 1NF. Which is the highest Normal Form (NF) it violates (2NF or 3NF)? Justify your answer by giving an example of a dependency violation.
C. Decompose the table to Second Normal Form (2NF). List the new tables created and state the specific dependencies that were removed.
D. Decompose the 2NF tables further to Third Normal Form (3NF). List all final tables in their 3NF form with their PK and FK
E. In the original FILM_CASTING table, AgentPhone is stored repeatedly for every film an actor is in. Which type of data redundancy anomaly (Insert, Update, or Delete) would this cause if the agent's phone number changes?

Question#5: Examine the **City College Enrollment Form** shown below:

City College

ENROLLMENT FORM

Student ID: S1001 **First Name:** James

Last Name: Smith **Address:** 123 Oak Street

Course ID	Course Name	Credits	Instructor Name	Enrollment Date
CS101	Introduction to Computer Science	3	Dr. Baker	15/01/2024
MA202	Calculus II	4	Dr. Johnson	15/01/2024
PH301	Physics	3	Dr. Lee	15/01/2024

1. Identify the **functional dependencies** represented by the attributes shown in the form above. State any assumptions that you make about the data and the attributes.

2. Describe and illustrate the process of **normalizing** the attributes shown in the form above to produce a set of well-designed **3NF (Third Normal Form)** relations.

3. Identify the **primary**, **alternate**, and **foreign keys** in your 3NF relations.

Good Luck!